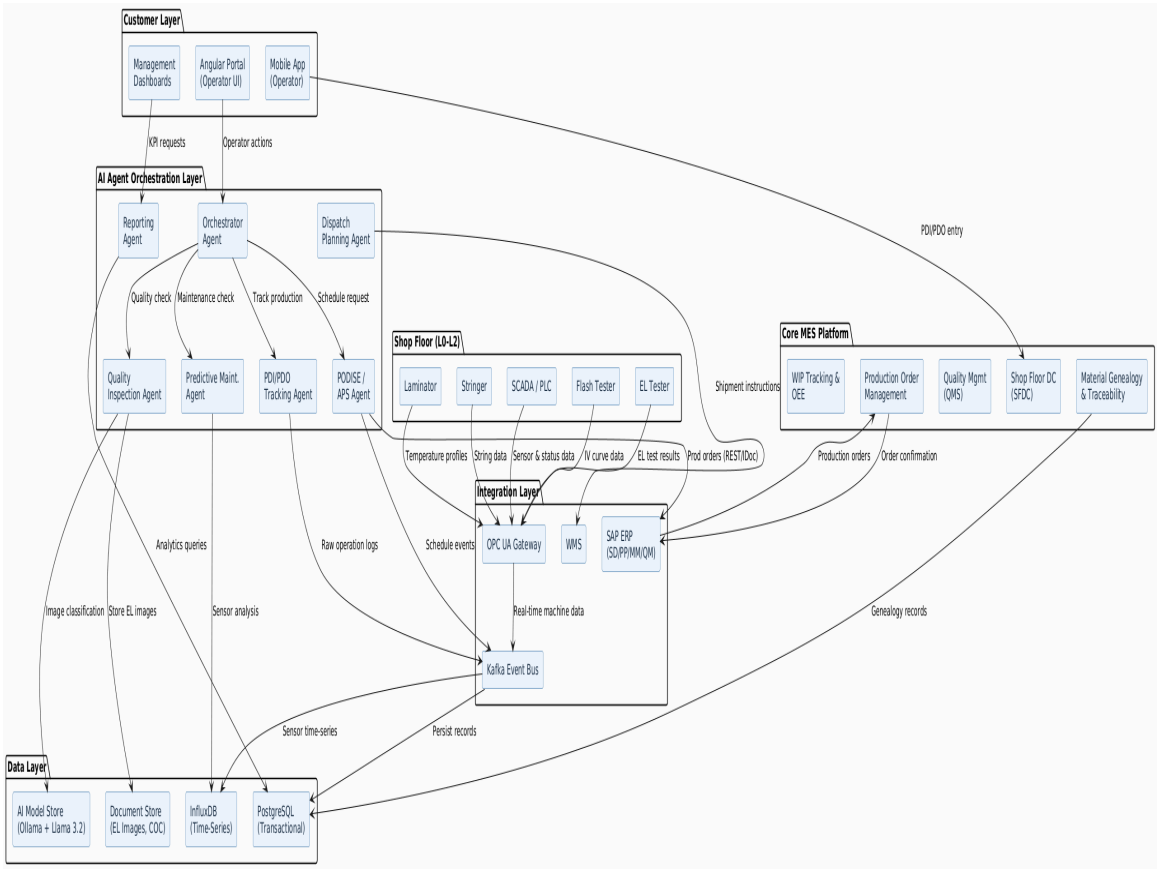
High-Level Design Document

Solar MES — AI-Native Architecture

Project: MES Implementation — Solar Manufacturing | Date: 02 May 2026

1. System Architecture Overview

The Solar MES is an AI-native, event-driven manufacturing execution platform. It integrates SAP ERP with shop-floor machines via an intelligent multi-agent orchestration layer, providing real-time production tracking, ALMM-compliant traceability, AI-powered quality inspection, and intelligent scheduling through PODISE.



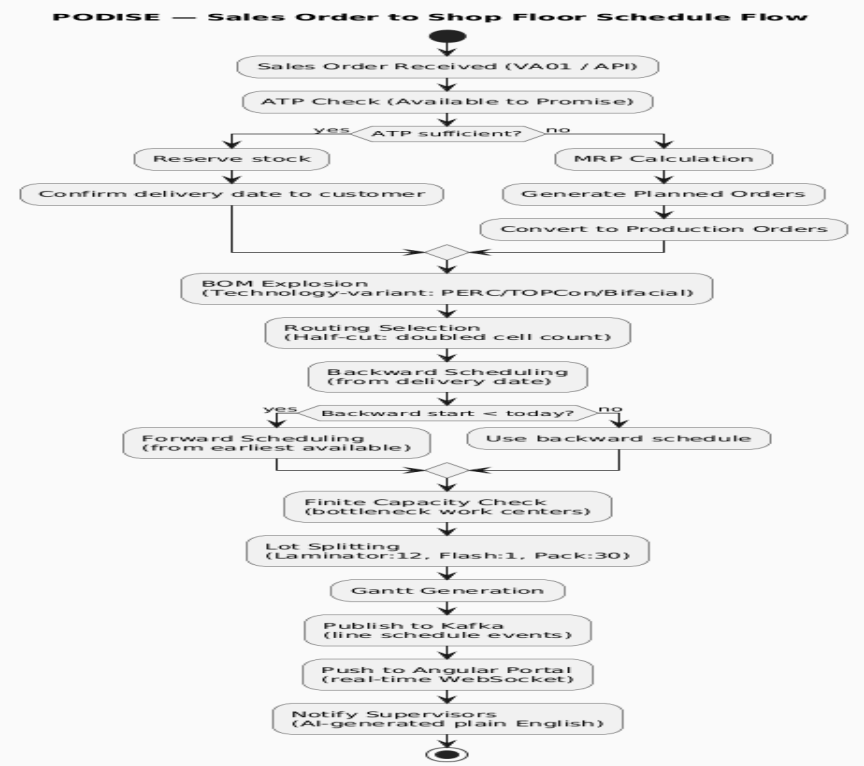
2. System Components

Component	Role	Technology
SAP ERP (SD/PP/MM/QM)	Production orders, BOM, finance, procurement	SAP S/4HANA or ECC
Solar MES Platform	Core execution: order mgmt, SFDC, QMS, OEE, traceability	Java Spring Boot + Kafka

PODISE / APS Engine	Real-time sales→production order conversion; lot splitting; scheduling	Java + Temporal.io + Ollama
Line Scheduler	Finite-capacity Gantt; WebSocket push to UI; batch-aware	Java + Angular
PDI/PDO Tracking Agent	OPC UA/MQTT/Excel ingest → Kafka → PostgreSQL; AI insights	Python + LangChain4j + Kafka
Quality Inspection Agent	EL/AOI image classification; flash test capture; defect coding	Python + CNN + Ollama
Predictive Maint. Agent	Sensor stream analysis; failure forecast; maintenance WO creation	Python + InfluxDB + Ollama
OPC UA Gateway	L2↔L3 machine data bridge; tag mapping; subscription model	Kepware / Ignition / machine-native
Kafka Event Bus	Event backbone; decoupling; replay; horizontal scaling	Apache Kafka + Zookeeper
PostgreSQL	Transactional data; audit trail; workflow state; genealogy	PostgreSQL 15+
InfluxDB	Time-series sensor data; OEE metrics; trend analysis	InfluxDB 2.x
Angular Portal	Operator UI; Gantt; OEE dashboards; quality alerts; shift reports	Angular 17 + WebSocket
AI Inference (Ollama)	On-premises LLM; Llama 3.2; image classification; NL reports	Ollama + GPU server
MCP Tool Registry	Uniform tool interface for agents to call SAP, Kafka, OPC UA, JDBC	Model Context Protocol
WMS	Inbound/outbound; inventory; LPN scanning; dock assignment	SAP WM/EWM or standalone

3. PODISE — Production Order Design & Intelligent Scheduling

PODISE solves the fundamental limitation of SAP MRP for solar manufacturing: it converts sales orders to production orders in under 2 seconds, handles solar-specific lot-size constraints (laminator batches of 4-12, flash tester 1, packaging 30), and re-optimizes schedules within 30 seconds of any disruption.

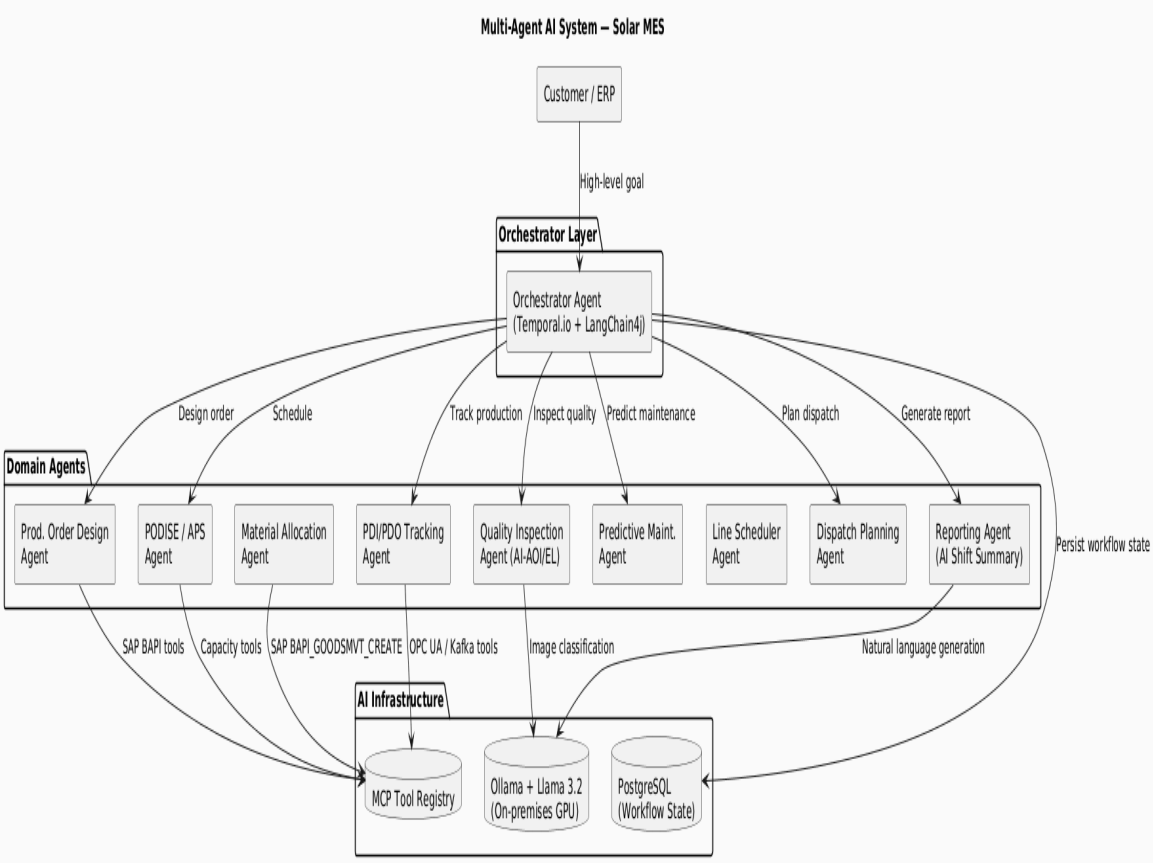


APS vs. SAP MRP — Capability Comparison

Capability	SAP MRP	PODISE/APS
Scheduling granularity	Day-level	Hour / minute-level
Capacity handling	Infinite (ignores constraints)	Finite (real constraints respected)
Rush order response	Batch MRP run required (1+ hour)	< 30 seconds automated
Lot-size constraints	Basic min/max	Complex multi-level (solar-specific)
Technology variants	Limited BOM handling	Full PERC/TOPCon/Bifacial/HJT
Bottleneck visibility	None	Real-time bottleneck identification
Re-optimization on breakdown	Manual replanning	Automatic on Kafka event
Gantt visualization	None built-in	Real-time Angular Gantt; drag-drop
AI recommendations	None	Plain-English rescheduling advice

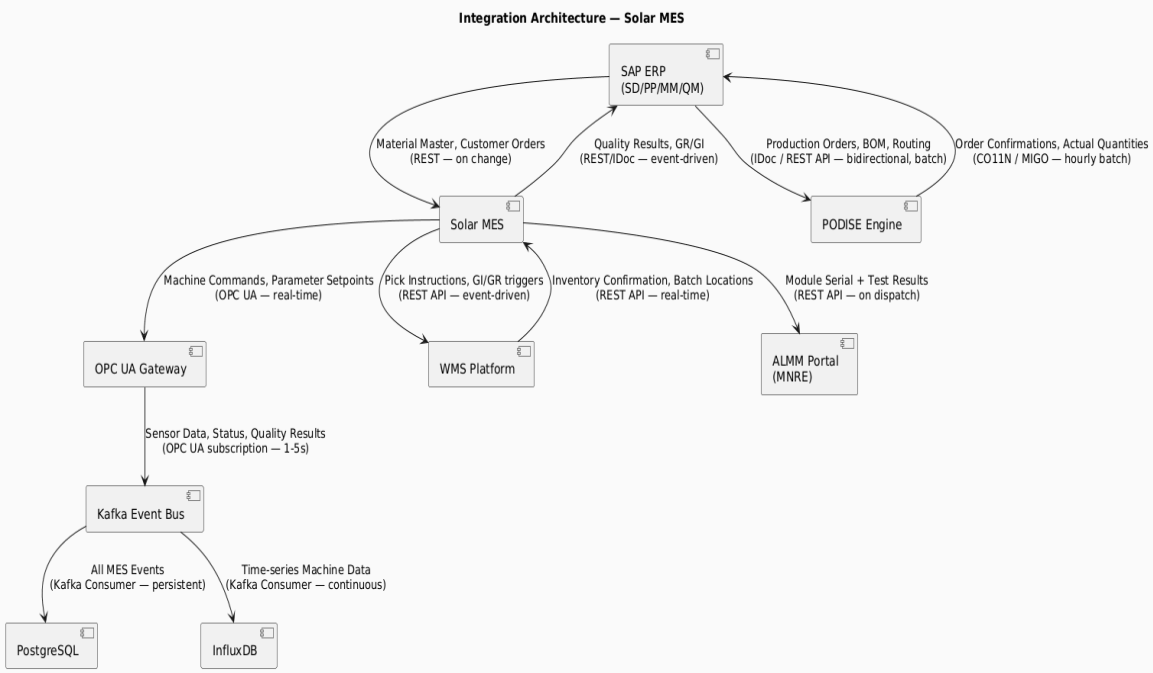
4. Multi-Agent AI System Design

The MES orchestrates 9 specialized AI agents coordinated by a supervisor agent (LangChain4j @SupervisorAgent pattern). Each agent has a specific domain, communicates via Kafka, and uses the MCP tool registry to call external services. The on-premises Ollama + Llama 3.2 ensures manufacturing IP stays within the factory.



Agent	Primary Responsibility	Key AI Capability
Orchestrator Agent	Receives high-level goals; coordinates all agents; persists workflow state in PostgreSQL	Goal decomposition; fault-tolerant task sequencing
PODISE / APS Agent	Sales order → production order in <2s; lot splitting; backward/forward scheduling	Constraint-based scheduling; multi-objective optimization
Line Scheduler Agent	Finite-capacity scheduling per work center; Gantt push via WebSocket	Real-time re-optimization on breakdown or rush order
PDI/PDO Tracking Agent	Ingests machine data (OPC UA, MQTT, Excel); publishes Kafka events; PDI/PDO records	AI scrap analysis; cycle-time anomaly detection; NL recommendations
Quality Inspection Agent	AI-AOI + EL image classification; flash test capture; defect coding and disposition	CNN defect detection >98% accuracy; IEC 61215 compliance
Predictive Maint. Agent	Monitors sensor streams; forecasts machine failure hours ahead; creates PM work orders	Time-series anomaly detection; MTBF prediction
Material Allocation Agent	Auto-reserves components in SAP; manages shortages; supplier alerts	Shortage prediction; alternative material suggestion
Dispatch Planning Agent	AI-optimized pallet/truck loading; dock assignment; ALMM COC generation	Route optimization; fill-rate maximization
Reporting Agent	Shift-end summaries; OEE trends; scrap analysis; plain-English AI narration	LLM-generated natural language reports for management

5. Integration Architecture



Integration	Direction	Protocol / Method	Frequency	Key Data
SAP ERP ↔ PODISE	Bidirectional	REST API + IDoc	Real-time / Hourly batch	Sales orders, prod orders, BOM, routings, confirmations
MES ↔ SAP (QM/WM)	Bidirectional	REST API / IDoc	Event-driven	Quality results, GR/GI, material movements
OPC UA Gateway → Kafka	Unidirectional	OPC UA subscription	1-5 seconds	Machine status, sensor values, alarm events
Flash Tester → MES	Unidirectional	OPC UA / REST	Per test cycle	IV curve, Pmax, Voc, Isc, FF, temperature
EL Tester → MES	Unidirectional	OPC UA / File	Per test cycle	Image classification result, defect type, coordinates
MES ↔ WMS	Bidirectional	REST API	Event-driven	Pick instructions, inventory confirmation, batch locations
MES → ALMM Portal	Unidirectional	REST API (MNRE)	On dispatch	Module serial, test results, COC data
Kafka → PostgreSQL	Unidirectional	Kafka Consumer	Continuous	All MES events (replay-capable audit trail)
Kafka → InfluxDB	Unidirectional	Kafka Consumer	Continuous	Time-series sensor data for OEE/trend analysis

6. End-to-End Data Flow

Step	From	To	Data	Storage
1	Machine (Stringer/Laminator)	OPC UA Gateway	Raw sensor readings, status, counts	Edge buffer
2	OPC UA Gateway	Kafka topic: raw-machine-data	Timestamped tag values	Kafka (7-day retention)
3	Kafka	PDI/PDO Tracking Agent	Operation log events	In-memory processing
4	PDI/PDO Agent	PostgreSQL	Validated PDI/PDO records, alerts	Permanent (audit trail)
5	PDI/PDO Agent	InfluxDB	Time-series metrics (OEE components)	90-day hot; S3 archive
6	EL Tester	Quality Inspection Agent	Raw EL image (TIFF/PNG)	GPU processing
7	Quality Inspection Agent	PostgreSQL + Document Store	Classification result + image path	Permanent (linked to serial)
8	Flash Tester	MES QMS Module	IV curve data per module serial	PostgreSQL (ALMM)
9	MES	SAP ERP	Production confirmation, actual quantities	SAP OLTP
10	MES	ALMM Portal (MNRE)	Module serial + all test results + COC	MNRE compliance DB
11	PostgreSQL + InfluxDB	Analytics Engine	Aggregated KPIs, trends	Data warehouse
12	Analytics Engine	Angular Portal	OEE gauges, Gantt, shift reports	Browser (WebSocket)

7. Deployment Architecture

Hybrid deployment model: Core MES, PODISE, and OPC UA integration run on-premise to ensure low latency (<100ms) for real-time shop-floor operations. Analytics, document storage, and AI model serving run on-premise GPU server. Cloud is optional for analytics dashboards and remote monitoring.

Layer	Component	Location	Hardware Spec
Application	Solar MES Platform	On-premise DC	2× 16-core server, 64 GB RAM, 2 TB SSD
Application	PODISE / APS Engine	On-premise DC	Shared with MES server
AI Inference	Ollama + Llama 3.2 + EL Classifier	On-premise GPU server	NVIDIA A10/RTX 4090, 48 GB VRAM

Database	PostgreSQL (primary)	On-premise DC	32-core, 128 GB RAM, 10 TB NVMe RAID-10
Database	InfluxDB (time-series)	On-premise DC	16-core, 64 GB RAM, 5 TB SSD
Messaging	Apache Kafka Cluster (3 nodes)	On-premise DC	3× 8-core, 32 GB RAM, 2 TB each
Edge	OPC UA Gateway per line	Shop floor (per line)	Industrial IPC (Advantech/Siemens), fanless
Network	OT/IT Segregated Network	Factory	Firewall DMZ; TLS 1.3; VLAN per line
Frontend	Angular Portal (Web)	On-premise app server	Shared; CDN for static assets
Display	OEE Display per Line	Shop floor (per line)	55" industrial display; Raspberry Pi / IPC

8. Security Architecture

Authentication & Authorisation: OAuth2 / JWT tokens; Role-Based Access Control (RBAC); per-module permissions; session timeout 8 hours

Network Security: OT/IT network segregation; DMZ for MES; TLS 1.3 for all API communication; no direct internet access for OT layer

Data Encryption: Encryption at rest (AES-256) for PostgreSQL and document store; encryption in transit (TLS) for all Kafka, REST, OPC UA traffic

Audit Logging: Immutable audit log for all user actions; retained 3 years minimum; ALMM compliance requirement

Backup & Recovery: Daily automated backup to encrypted off-site storage; RTO ≤ 4 hours; RPO ≤ 1 hour; quarterly DR drill

OT Security: OPC UA authentication (X.509 certificates); machine whitelist; no direct internet from PLC/SCADA layer

AI Data Privacy: On-premises LLM inference; manufacturing IP never sent to cloud APIs; model updates via air-gapped process

Vulnerability Management: Quarterly security assessment; CVE monitoring; patch management SLA (critical: 48h, high: 7 days)
